```
clc;
clear;
close all;

% Experiment 5: FIR Filter Design Comparison

Fs = 10000;
fp = 1500;
fst = 2000;
N = 40;

% Normalized frequency edges
F = [0 fp fst Fs/2] / (Fs/2);
A = [1 1 0 0];

% Parks-McClellan FIR filter
b_pm = firpm(N, F, A);

% Frequency-sampling FIR filter
b_fs = fir2(N, F, A);

% Display results
fprintf('EXPERIMENT 5: FIR FILTER DESIGN\n');
```

```
EXPERIMENT 5: FIR FILTER DESIGN
```

```
fprintf('Sampling Frequency = %d Hz\n', Fs);
```

```
Sampling Frequency = 10000 Hz
```

```
fprintf('Passband Edge = %d Hz\n', fp);
```

```
Passband Edge = 1500 Hz
```

```
fprintf('Stopband Edge = %d Hz\n', fst);
```

```
Stopband Edge = 2000 Hz
```

```
fprintf('Filter Order = %d\n\n', N);
```

```
Filter Order = 40
```

```
fprintf('Parks-McClellan Coefficients (First 10):\n');
```

```
Parks-McClellan Coefficients (First 10):
```

```
disp(b_pm(1:10));
```

```
   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037    0.0142    0.0100   -0.0090
```

```
fprintf('Frequency-Sampling Coefficients (First 10):\n');
```

Frequency-Sampling Coefficients (First 10):

```
disp(b_fs(1:10));
```

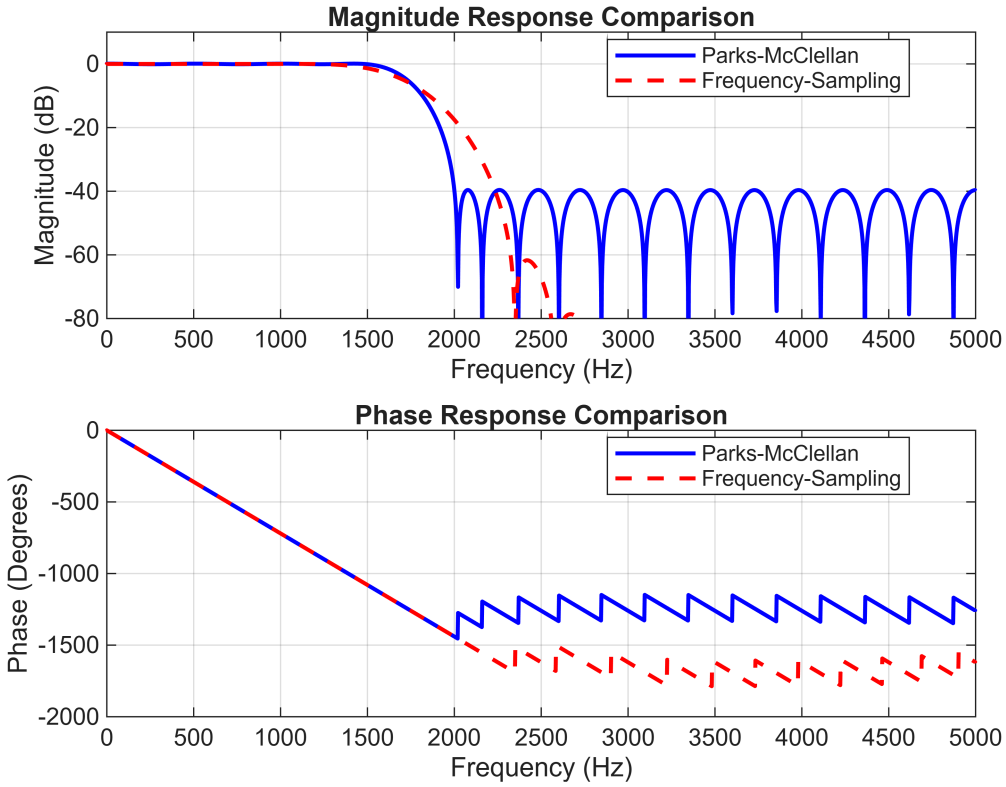
```
    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008    0.0035    0.0029   -0.0039
```

```
% Calculate frequency responses
[H_pm, f] = freqz(b_pm, 1, 2048, Fs);
[H_fs, ~] = freqz(b_fs, 1, 2048, Fs);

% Plot magnitude responses
figure('Name', 'FIR Filter Comparison');

subplot(2,1,1);
plot(f, 20*log10(max(abs(H_pm), 1e-6)), 'b', ...
    'LineWidth', 1.5);
hold on;
plot(f, 20*log10(max(abs(H_fs), 1e-6)), 'r--', ...
    'LineWidth', 1.5);
grid on;
legend('Parks-McClellan', 'Frequency-Sampling', ...
    'Location', 'best');
title('Magnitude Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xlim([0 Fs/2]);
ylim([-80 10]);

% Plot phase responses
subplot(2,1,2);
plot(f, unwrap(angle(H_pm))*180/pi, 'b', ...
    'LineWidth', 1.5);
hold on;
plot(f, unwrap(angle(H_fs))*180/pi, 'r--', ...
    'LineWidth', 1.5);
grid on;
legend('Parks-McClellan', 'Frequency-Sampling', ...
    'Location', 'best');
title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
xlim([0 Fs/2]);
```


Magnitude Response Comparison
Parks-McClellan
Frequency-Sampling
Magnitude (dB)
0
-20
-40
-60
-80
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)
Phase Response Comparison
Parks-McClellan
Frequency-Sampling
Phase (Degrees)
0
-500
-1000
-1500
-2000
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)